IQRASEC ACADEMY

Passive Reconnaissance Report

WAPT Phase 1

Task 2 Reconnaissance and Enumeration

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Learning platform

TryHackMe

Description:

This report covers public information gathering in the Passive Reconnaissance room. It explains the techniques, provides practical commands and searches, and records the evidence needed to support findings as the room is completed.

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1. Objective

To collect domain registration, DNS, subdomain and indexed service information; distinguish passive information gathering from direct probing; and interpret findings without treating public records as proof of a vulnerability.

2. Concept and Understanding

2.1 Passive and Active Reconnaissance

Passive reconnaissance gathers information from existing public records or third-party indexes. Active reconnaissance directly interacts with a target, for example by pinging it, scanning ports or testing web endpoints. The distinction depends on where requests go, rather than simply on the tool name.

A query to a public recursive DNS resolver does not directly probe the target web application, although a cache miss may cause the resolver to contact authoritative DNS servers. Passive collection is not necessarily invisible or free from logging.

2.2 WHOIS and RDAP

WHOIS provides domain registration records where the service remains available. RDAP is the modern registration-data protocol using HTTPS and structured JSON. ICANN identifies RDAP as the definitive source for generic top-level domain registration data from 28 January 2025. Records may reveal dates, registrar, name servers and status; personal contact fields may be redacted.

2.3 DNS Records

Record	Information provided
A and AAAA	IPv4 and IPv6 addresses respectively.
MX	Mail exchange hosts and preference values.
NS	Authoritative name servers.
TXT	Text such as verification data or email-policy records.
CNAME	An alias pointing to another DNS name.
SOA	Zone administration information, including its serial number.

2.4 Public Indexes and Certificate Logs

DNSDumpster aggregates domain infrastructure information. Certificate Transparency logs can reveal names listed in issued certificates. Shodan indexes service banners and host metadata. Indexed entries are leads: names may be historical, service data may be outdated, and shared infrastructure may belong to a hosting or CDN provider.

3. Practical Work

Use TryHackMe's example domains for the following exercises. Record the lookup date, source and actual output. Terminal commands can be run in Kali Linux or an available AttackBox; the report does not assume which environment was used.

3.1 Domain Registration Lookup

```
whois tryhackme.com
```

Review the registrar, registration date, name servers and domain status. If WHOIS is unavailable or incomplete, use the RDAP-based lookup at https://lookup.icann.org/. Record which source provided the answer rather than copying dates from the room's sample output.

3.2 DNS Address and Mail Lookups

```
nslookup -type=A tryhackme.com 1.1.1.1
dig @1.1.1.1 tryhackme.com A
dig @1.1.1.1 tryhackme.com MX
dig @1.1.1.1 tryhackme.com NS
```

Compare the A-record answers from nslookup and dig. In dig, inspect the response status, answer section, record type, TTL and resolver. For MX records, smaller preference numbers have higher priority. An address returned by DNS may belong to shared infrastructure and does not prove exclusive target ownership.

3.3 TXT Record Exercise

```
dig @1.1.1.1 thmlabs.com TXT
nslookup -type=TXT thmlabs.com 1.1.1.1
```

Inspect the returned TXT records to locate the training information requested by Task 4. TXT records contain text; not every TXT record is an email-security policy. Record the observation without placing answer flags in the public report.

If a lookup fails, record the actual error and the troubleshooting step used. A timeout does not establish that the requested DNS record is absent.

3.4 Subdomain Discovery with DNSDumpster

Open https://dnsdumpster.com/ and search for tryhackme.com. Review the host records, relationships and service information returned. For the room question, inspect the Services or Banners summary and record which category has the highest count.

3.5 Certificate Transparency Search

Open https://crt.sh/ and enter the following search:

```
%.tryhackme.com
```

Review certificate names and validity dates, remove duplicate hostnames from your notes, and distinguish explicit names from wildcard entries. A certificate record shows issuance history; it does not prove that a hostname currently resolves or is accessible.

3.6 Indexed Service Information with Shodan

Open https://www.shodan.io/ and search existing indexed records using:

```
hostname:tryhackme.com
```

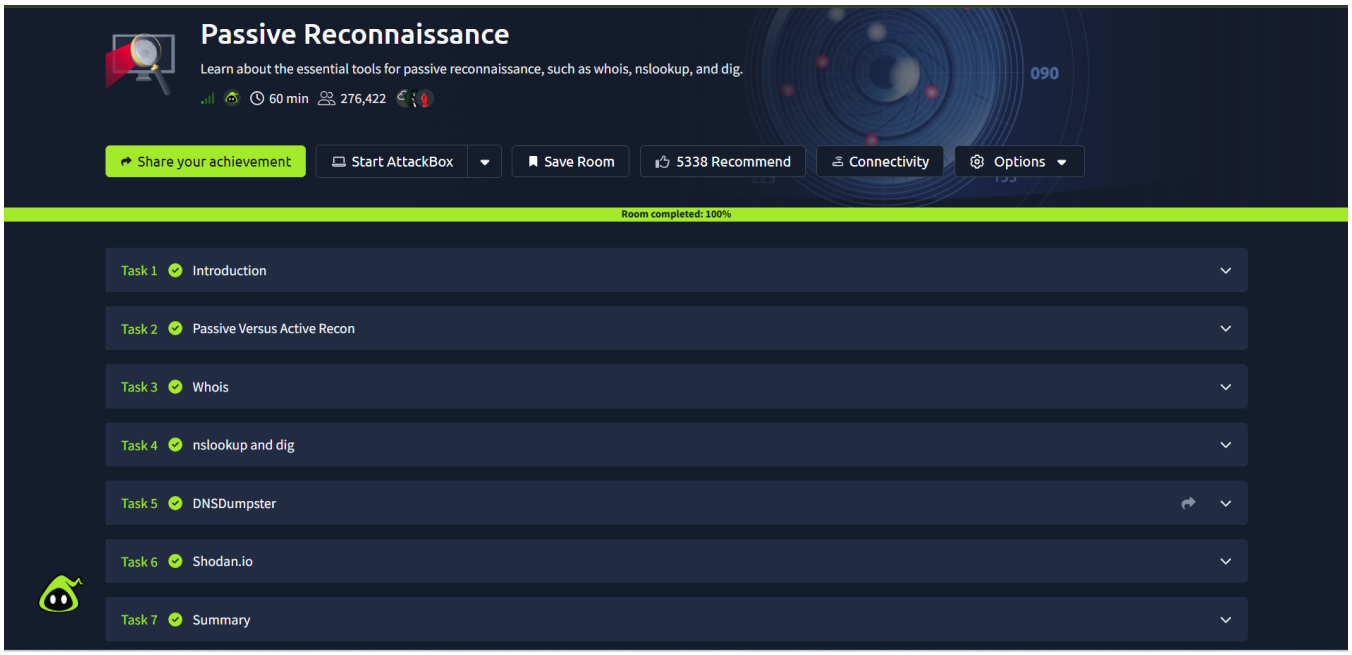
Review the displayed hostname, address, provider, service ports, banner and observation time. If using a DNS-derived address, note that shared CDN addresses may represent multiple customers. Do not infer that every indexed service belongs to TryHackMe.

4. Tools Used

Tool or source	Purpose
WHOIS or ICANN Lookup	Retrieve available domain registration data.
nslookup and dig	Query DNS records through a specified resolver.
DNSDumpster and crt.sh	Review domain infrastructure and certificate names.
Shodan	Review previously indexed service information.
Terminal and web browser	Run commands and access public lookup interfaces.

5. Evidence

Capture your commands or searches together with readable results. Insert the images with the figure captions below. Add more than one image when an output or summary cannot fit clearly into a single screenshot.



6. Key Learnings

Public records reveal different parts of a domain's infrastructure. Registration data identifies administrative details; DNS describes naming and service routing; certificates suggest candidate names; and indexed banners provide historical service observations. Findings become stronger when sources agree and timestamps are retained. Direct validation of services belongs to the subsequent active reconnaissance stage within its agreed scope.

8. References

TryHackMe. Passive Reconnaissance. https://tryhackme.com/room/passiverecon

ISC. BIND 9 manual pages for dig and nslookup. https://bind9.readthedocs.io/en/stable/manpages.html

ICANN. Registration Data Access Protocol information. https://www.icann.org/en/contracted-parties/registry-operators/registration-data-access-protocol/information-for-rdap-users-31-08-2018-en

Shodan. Search query fundamentals. https://help.shodan.io/the-basics/search-query-fundamentals